

10-9-2023

MATHEMATICS

classmate

Date

Page

1. (1) 3
2. (4) No solution
3. (1) 10 units
4. (2) proportional
5. (2) $-\frac{1}{2}$
6. (1) x^3y^3

$$\sqrt{3} \tan \theta = 2 \sin \theta$$

$$\sqrt{3} \frac{\sin \theta}{\cos \theta} = 2 \sin \theta$$

$$\sqrt{3} \sec \theta = 2$$

$$\sec \theta = \frac{2}{\sqrt{3}}$$

$$\cos \theta = \frac{\sqrt{3}}{2}$$

$$\theta = 30^\circ$$

$$\sin 30^\circ = \cos^2 30^\circ$$

$$\left(\frac{1}{2}\right)^2 = \left(\frac{\sqrt{3}}{2}\right)^2$$

$$\frac{1}{4} - \frac{3}{4} = \frac{1-3}{4} = \frac{-2}{4} = \frac{-1}{2}$$

$$x^2 - 2x - (7p+3) = 0$$

$$(-4)^2 - 2(-4) - 7p+3 = 0$$

$$16 + 8 - 7p+3 = 0$$

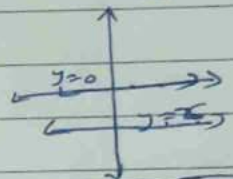
$$24+3 - 7p = 0$$

$$27 - 7p = 0$$

$$-7p = -27$$

$$p = \frac{27}{7}$$

$$p = 3$$



$$\sqrt{(6-0)^2 + (4-0)^2}$$

$$\sqrt{(-6)^2 + (8)^2}$$

$$\sqrt{36 + 64}$$

$$\sqrt{100}$$

$$= 10$$

7. Let α and β be the roots of the polynomial $p(x) = 4x^2 - 2x + (k-4)$, where, $a=4$, $b=-2$, and $c=k-4$. According to the question,

$$\alpha + \beta = \frac{1}{2} \alpha \beta$$

$$\frac{-b}{a} = \frac{1}{2} \frac{c}{a}$$

$$\frac{-(-2)}{4} = \frac{1}{2} \left(\frac{k-4}{4} \right)$$

$$\frac{2}{4} = \frac{1}{2} \left(\frac{k-4}{4} \right)$$

$$\frac{2 \times 2}{4 \times 1} = \frac{k-4}{4} \Rightarrow \frac{k-4}{4} = 1 \Rightarrow k-4 = 4$$

$$\Rightarrow k = 4 + 4 = 8 \Rightarrow \boxed{k=8}$$

$\therefore$ ~~and~~ The value of K is 8

8. Let the points $A(0,0)$, $B(3,\sqrt{3})$ and $C(3,x)$ form an equilateral triangle ABC

$$\therefore AB = BC = AC$$

$$\begin{aligned} AB &= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \\ &= \sqrt{(0 - 3)^2 + (0 - \sqrt{3})^2} \\ &= \sqrt{(-3)^2 + (-\sqrt{3})^2} \\ &= \sqrt{9 + 3} \\ &= \sqrt{12} \\ &= 2\sqrt{3} \text{ units} \end{aligned}$$

$$\begin{aligned} AC &= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \\ &= \sqrt{(0 - 3)^2 + (0 - x)^2} \\ &= \sqrt{(-3)^2 + (-x)^2} \\ &= \sqrt{9 + x^2} \text{ units} \end{aligned}$$

$$AC = AB$$

$$\sqrt{9 + x^2} = 2\sqrt{3}$$

Squaring on both sides,

$$9 + x^2 = 12$$

$$x^2 = 12 - 9$$

$$x^2 = 3$$

$$x = \sqrt{3}$$

$$\boxed{x = \pm\sqrt{3}}$$

$\therefore$ The value of x is $+\sqrt{3}$ or $-\sqrt{3}$



9.

$$2px + 3y - 7 = 0 \rightarrow \text{①}$$

$$2x + y - 6 = 0 \rightarrow \text{②}$$

$$a_1 = 2p \quad ; \quad b_1 = 3 \quad ; \quad c_1 = -7$$

$$a_2 = 2 \quad ; \quad b_2 = 1 \quad ; \quad c_2 = -6$$

According to the questions,

$$\frac{a_1}{a_2} \neq \frac{b_1}{b_2} \quad (\text{unique solution})$$

$$\frac{2p}{2} \neq \frac{3}{1}$$

$$p \neq 3$$

$$\therefore p \neq 3$$

$\therefore p$ can have any real value except 3

10. Given:- $PD \parallel BC$

$$\& \frac{AD}{DC} = \frac{CE}{BE}$$

To prove:- $PE \parallel AC$

Proof:-

In $\triangle ABC$,

$PD \parallel BC$ (given)

$$\therefore \frac{AP}{PB} = \frac{AD}{DC} \quad (\text{By BPT})$$

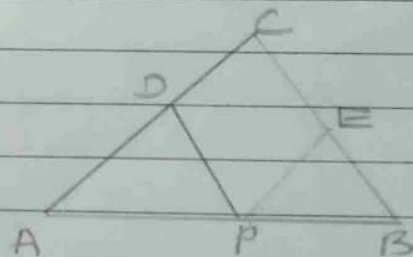
$$\frac{AD}{DC} = \frac{CE}{BE} \rightarrow \text{②}$$

From ① & ② we get,

$$\frac{AP}{PB} = \frac{CE}{BE} \Rightarrow \frac{PB}{AP} = \frac{BE}{CE}$$

$\therefore PE \parallel AC$ (By converse of BPT)

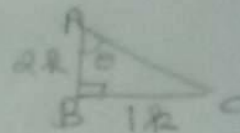
Hence, proved.



$$11. \quad \tan \theta = \frac{1}{2} = \frac{\text{Opp.}}{\text{adj.}}$$

Let, the opposite and adjacent sides to θ be $1k$ and $2k$ respectively where k is any real number.

$$\begin{aligned} AC^2 &= AB^2 + BC^2 \quad (\text{By Pythagoras theorem}) \\ AC &= \sqrt{AB^2 + BC^2} \\ &= \sqrt{(2k)^2 + (1k)^2} \\ &= \sqrt{4k^2 + 1k^2} \\ &= \sqrt{5k^2} \end{aligned}$$



$$\text{Hypotenuse (AC)} = \sqrt{5}k$$

$$\sin \theta = \frac{\text{Opp.}}{\text{Hyp.}} = \frac{1k}{\sqrt{5}k} = \frac{1}{\sqrt{5}}$$

$$\cos \theta = \frac{\text{Adj.}}{\text{Hyp.}} = \frac{2k}{\sqrt{5}k} = \frac{2}{\sqrt{5}}$$

$$\begin{aligned} \text{LHS} &:- \frac{1}{2} \sin \theta \cos \theta \\ &= \frac{1}{2} \left(\frac{1}{\sqrt{5}} \right) \left(\frac{2}{\sqrt{5}} \right) \\ &= \frac{1}{5} \end{aligned}$$

$$\text{RHS} :- \frac{1}{5}$$

$$\text{LHS} = \text{RHS}$$

Hence proved

12.

$$\frac{257}{5000} = \frac{257}{2^3 \times 5^4} \quad (2^n = 2^3 \text{ and } 5^m = 5^4)$$

$$= \frac{257 \times 2}{2^3 \times 5^4 \times 2} = \frac{514}{10000}$$

$$= \boxed{0.0514}$$

$$\begin{array}{r} 0.0514 \\ 257 \\ \times 2 \\ \hline 514 \end{array}$$

$$2^3 \times 5^4$$

13. Let $\sqrt{3} = \frac{p}{q}$, where p and q are co-prime and

$q \neq 0$

Squaring ~~both~~ on both sides,

$$(\sqrt{3})^2 = \left(\frac{p}{q}\right)^2$$

$$3 = \frac{p^2}{q^2}$$

$$3q^2 = p^2 \rightarrow \textcircled{1}$$

p^2 is divisible by 3

p is also divisible by 3

3 is a factor of p

Let, $p = 3c$

Squaring on both sides

$$(p)^2 = (3c)^2$$

$$p^2 = 9c^2 \rightarrow \textcircled{2}$$

From $\textcircled{1}$ & $\textcircled{2}$ we get,

$$3q^2 = 9c^2$$

$$q^2 = 3c^2$$

q^2 is divisible by 3

q is also divisible by 3

3 is a factor of q

$\therefore$ 3 is a factor of both p and q

It is a contradiction

Our assumption is wrong

$\therefore \sqrt{3}$ is irrational number.

[Let p be a prime number. If p divides a^2 then p divides a , where a is a positive integer]

[Let p be a prime number. If p divides a^2 then p divides a , where a is a positive integer]

14.

$$p(x) = 2x^2 - 3x + 7$$

$$\alpha + \beta = \frac{-b}{a} = \frac{-(-3)}{2} = \frac{3}{2}$$

$$\alpha\beta = \frac{c}{a} = \frac{7}{2}$$

$$\begin{aligned} \frac{1}{2\alpha-3} + \frac{1}{2\beta-3} &= \frac{2\beta-3+2\alpha-3}{(2\alpha-3)(2\beta-3)} \\ &= \frac{2(\alpha+\beta)-6}{4\alpha\beta-6\alpha-6\beta+9} \\ &= \frac{2(\alpha+\beta)-6}{4\alpha\beta-6(\alpha+\beta)+9} \\ &= \frac{2\left(\frac{3}{2}\right)-6}{2\cdot 4\left(\frac{7}{2}\right)-3\cdot 6\left(\frac{3}{2}\right)+9} \\ &= \frac{3-6}{14-9+9} \\ &= \boxed{\frac{-3}{14}} \end{aligned}$$

$$15. \quad \begin{aligned} 3x - (a+1)y - (2b-1) &= 0 \rightarrow (1) \\ 5x + (1-2a)y - 3b &= 0 \rightarrow (2) \end{aligned}$$

For infinitely many solutions,

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$

$$\frac{3}{5} = \frac{-(a+1)}{1-2a} = \frac{-(2b-1)}{-3b}$$

$$\frac{3}{5} = \frac{-a-1}{1-2a}$$

$$3-6a = -5a-5$$

$$5+3 = -5a+6a$$

$$8 = a$$

$$\boxed{a=8}$$

$$\frac{3}{5} = \frac{2b-1}{3b}$$

$$9b = 10b-5$$

$$9b-10b = -5$$

$$-b = -5$$

$$\boxed{b=5}$$

$$\therefore a=8 \text{ \& } b=5$$

16. Given:- $\triangle ABE \cong \triangle ACD$

To prove:- $\triangle ADE \sim \triangle ABC$

Proof:-

$\triangle ABE \cong \triangle ACD$ (given)

$$AC = AB \rightarrow \textcircled{1} \quad \left. \begin{array}{l} AC = AB \rightarrow \textcircled{1} \\ AE = AD \rightarrow \textcircled{2} \end{array} \right\} \text{(CPCT)}$$

$$AE = AD \rightarrow \textcircled{2}$$

$$\textcircled{1} \div \textcircled{2}$$

$$\frac{AC}{AE} = \frac{AB}{AD} \Rightarrow \frac{AE}{AC} = \frac{AD}{AB}$$

In $\triangle ADE$ & $\triangle ABC$,

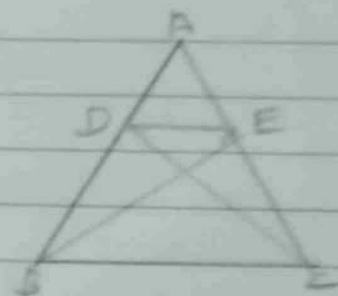
$$\angle DAE = \angle BAC \text{ (common)}$$

$$\frac{AE}{AC} = \frac{AD}{AB} \text{ (Proved above)}$$

$\therefore$ By SAS Similarity criteria

$$\triangle ADE \sim \triangle ABC$$

Hence, proved.



$$17. \text{ LHS :- } \frac{\tan^2 \theta}{\tan^2 \theta - 1} + \frac{\operatorname{cosec}^2 \theta}{\sec^2 \theta - \operatorname{cosec}^2 \theta}$$

$$\frac{\cancel{\tan^2 \theta}}{\cancel{\tan^2 \theta} - \cancel{\sec^2 \theta} \tan^2}$$

$$= \frac{\frac{\sin^2 \theta}{\cos^2 \theta}}{\frac{\sin^2 \theta - \cos^2 \theta}{\cos^2 \theta}} + \frac{\frac{1}{\sin^2 \theta}}{\frac{1}{\cos^2 \theta} - \frac{1}{\sin^2 \theta}}$$

$$= \frac{\sin^2 \theta \times \cos^2 \theta}{\cos^2 \theta \sin^2 \theta - \cos^2 \theta} + \frac{\frac{1}{\sin^2 \theta}}{\frac{\sin^2 \theta - \cos^2 \theta}{\sin^2 \theta \cos^2 \theta}}$$

$$= \frac{\sin^2 \theta}{\sin^2 \theta - \cos^2 \theta} + \frac{1}{\sin^2 \theta} \times \frac{\sin^2 \theta \cos^2 \theta}{\sin^2 \theta - \cos^2 \theta}$$

$$= \frac{\sin^2 \theta}{\sin^2 \theta - \cos^2 \theta} + \frac{\cos^2 \theta}{\sin^2 \theta - \cos^2 \theta}$$

$$= \frac{\sin^2 \theta + \cos^2 \theta}{\sin^2 \theta - \cos^2 \theta}$$

$$= \frac{1}{\sin^2 \theta - \cos^2 \theta} \quad [\sin^2 \theta + \cos^2 \theta = 1]$$

$$\text{RHS :- } \frac{1}{\sin^2 \theta - \cos^2 \theta}$$

$$\text{LHS} = \text{RHS}$$

Hence, proved.

18. Point $C(1, 1)$ divides the line segment joining $A(-2, 7)$ and B internally in ratio $m_1 : m_2 = 3 : 2$

$$C(x_1) = \left(\frac{m_1 x_2 + m_2 x_1}{m_1 + m_2}, \frac{m_1 y_2 + m_2 y_1}{m_1 + m_2} \right)$$

$$(1, 1) = \left(\frac{(3 \times x_2) + (2 \times -2)}{5}, \frac{(3 \times y_2) + (2 \times 7)}{5} \right)$$

$$(b) = \left(\frac{3x_2 - 4}{5}, \frac{3y_2 + 14}{5} \right)$$

$$1 = \frac{3x_2 - 4}{5}$$

$$5 = 3x_2 - 4$$

$$5 + 4 = 3x_2$$

$$9 = 3x_2$$

$$\boxed{x_2 = 3}$$

$$1 = \frac{3y_2 + 14}{5}$$

$$5 = 3y_2 + 14$$

$$5 - 14 = 3y_2$$

$$-9 = 3y_2$$

$$\boxed{y_2 = -3}$$

$\therefore$ The coordinates of B are $(3, -3)$

19. Let the fraction be $\frac{x}{y}$

According to the question,

$$2\left(\frac{x}{y+1}\right) = 1$$

$$\frac{2x}{y+1} = 1$$

$$y+1$$

$$2x = y+1 \Rightarrow 2x - y = 1 \rightarrow \textcircled{1}$$

$$\frac{x+4}{2y} = \frac{1}{2}$$

$$2y$$

$$2$$

$$2y = 2x + 8 \quad (\div 2)$$

$$y = x + 4$$

$$-x + y = 4 \rightarrow \textcircled{2}$$

$$\textcircled{1} + \textcircled{2}$$

$$2x - y = 1$$

$$-x + y = 4$$

$$\boxed{x = 5}$$

Sub $x = 5$ in $\textcircled{1}$, we get,

$$2(5) - y = 1$$

$$10 - y = 1 \Rightarrow -y = 1 - 10 \Rightarrow -y = -9 \Rightarrow \boxed{y = 9}$$

$\therefore$ The function is $\frac{5}{9}$

20. Basic Proportionality Theorem?

The line passing through two sides of a triangle parallel to the third side, then, the other two sides of the triangle are divided in the same ratio.

Given:- In $\triangle ABC$ line l' passes parallel to the side BC through the points D and E



To prove:- $\frac{AD}{DB} = \frac{AE}{EC}$

Proof:-

Area of a triangle = $\frac{1}{2} \times \text{base} \times \text{height}$ 3V units

$$\text{ar}(\triangle ADE) = \frac{1}{2} \times AD \times EN \rightarrow (1)$$

$$\text{ar}(\triangle DEB) = \frac{1}{2} \times DB \times EN \rightarrow (2)$$

$$(1) \div (2)$$

$$\frac{\text{ar}(\triangle ADE)}{\text{ar}(\triangle DEB)} = \frac{\frac{1}{2} \times AD \times EN}{\frac{1}{2} \times DB \times EN}$$

$$\frac{\text{ar}(\triangle ADE)}{\text{ar}(\triangle DEB)} = \frac{AD}{DB} \rightarrow (3)$$

$$\text{ar}(\triangle ADE) = \frac{1}{2} \times AE \times DM \rightarrow (4)$$

$$\text{ar}(\triangle DEC) = \frac{1}{2} \times EC \times DM \rightarrow (5)$$

④ ÷ ⑤

$$\frac{\text{ar}(\triangle ADE)}{\text{ar}(\triangle DEC)} = \frac{\frac{1}{2} \times AE \times DM}{\frac{1}{2} \times EC \times DM}$$

$$\frac{\text{ar}(\triangle ADE)}{\text{ar}(\triangle DEC)} = \frac{AE}{EC} \rightarrow \textcircled{6}$$

$\triangle DEC$ and $\triangle DEB$ lie on the same base DE and between the same parallels DE and BC

$$\therefore \text{ar}(\triangle DEC) = \text{ar}(\triangle DEB) \rightarrow \textcircled{7}$$

From ③, ⑥ & ⑦

$$\frac{AD}{DB} = \frac{AE}{EC}$$

$$21. = \frac{\sin 30^\circ + 2\cos^2 45^\circ + \tan^2 60^\circ}{\frac{1}{2}\cot 45^\circ + \cos^2 30^\circ + \tan^2 45^\circ}$$

$$= \frac{\frac{1}{2} + 2\left(\frac{1}{\sqrt{2}}\right)^2 + (\sqrt{3})^2}{\frac{1}{2}(1) + \left(\frac{\sqrt{3}}{2}\right)^2 + (1)^2}$$

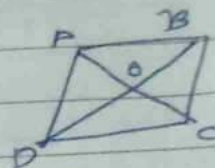
$$= \frac{\frac{1}{2} + 2\left(\frac{1}{2}\right) + 3}{\frac{1}{2} + \frac{3}{4} + 1}$$

$$= \frac{\frac{1}{2} + 1 + 3}{\frac{1}{2} + \frac{3}{4} + 1} = \frac{\frac{1}{2} + 4}{\frac{1}{2} + \frac{3}{4} + 1} = \frac{\frac{1+8}{2}}{\frac{2+3+4}{4}}$$

$$= \frac{9}{\frac{9}{4}} = 9 \times \frac{4}{9}$$

$$= 4$$

22. Let Three consecutive vertices of a parallelogram are $A(1,2)$, $B(2,3)$, $C(6,5)$ and $D(x,y)$



Diagonals of a parallelogram bisect each other

Let O be the point of their intersection

O is the midpoint of AC

$$O(x,y) = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

$$= \left(\frac{1 + 6}{2}, \frac{2 + 5}{2} \right)$$

$$O = \left(\frac{7}{2}, \frac{7}{2} \right)$$

O is the midpoint of BD

$$O\left(\frac{7}{2}, \frac{7}{2}\right) = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

$$\left(\frac{7}{2}, \frac{7}{2} \right) = \left(\frac{2 + x}{2}, \frac{3 + y}{2} \right)$$

$$\frac{7}{2} = \frac{2 + x}{2}$$

$$x = 7 - 2$$

$$\boxed{x = 5}$$

$$\frac{7}{2} = \frac{3 + y}{2}$$

$$y = 7 - 3$$

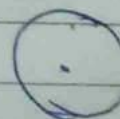
$$\boxed{y = 4}$$

$\therefore$ The coordinates of the vertex D is $(5,4)$

23. Speed of A = 330 m/min

Speed of B = 198 m/min

Speed of C = 220 m/min



$$\begin{array}{r} 3 \overline{) 330} \\ 2 \overline{) 165} \\ 5 \overline{) 33} \\ 11 \overline{) 11} \\ 1 \end{array}$$

$$\begin{array}{r} 2 \overline{) 198} \\ 3 \overline{) 99} \\ 3 \overline{) 33} \\ 11 \overline{) 11} \\ 1 \end{array}$$

$$\begin{array}{r} 2 \overline{) 220} \\ 2 \overline{) 110} \\ 5 \overline{) 55} \\ 11 \overline{) 11} \\ 1 \end{array}$$

$$330 = 2 \times 3 \times 5 \times 11$$

$$198 = 2 \times 3^2 \times 11$$

$$220 = 2^2 \times 5 \times 11$$

$$\text{LCM} = 2^2 \times 3^2 \times 5 \times 11$$

$$= 36 \times 5 \times 11$$

$$= 396 \times 5$$

$$= 1980 \text{ m/min}$$

$$v = \frac{d}{t}$$

$$t = \frac{d}{v} = \frac{1980}{1980} = 1 \text{ min}$$

$$t_A = \frac{1980}{330} = 6 \text{ min} \quad t_B = \frac{1980}{198} = 10 \text{ min} \quad t_C = \frac{1980}{220} = 9 \text{ min}$$

$$6 = 3 \times 2$$

$$10 = 5 \times 2$$

$$9 = 3 \times 3$$

$$\text{LCM} = 2 \times 5 \times 3^2$$

$$= 10 \times 9$$

$$= 90 \text{ min}$$

$\therefore$ They will meet together after 90 min

$$\begin{array}{r} 32 \\ \times 11 \\ \hline 320 \\ + 320 \\ \hline 352 \end{array}$$

$$\begin{array}{r} 1980 \\ \times 6 \\ \hline 11880 \end{array}$$

$$\begin{array}{l} 3 \times 2 \\ 5 \times 2 \\ 3 \times 3 \end{array}$$

24. let father's age be x years and son's age be y years

$$x + 2y = 70 \rightarrow (1)$$

$$2x + y = 95 \rightarrow (2)$$

$$(1) + (2) \Rightarrow 3x + 3y = 165$$

$$x + y = 55 \rightarrow (3)$$

$$x + 2y = 70 \rightarrow (1)$$

$$2x + y = 95 \rightarrow (2)$$

(1)

(2)

(1)

$$\begin{array}{r} 95 \\ -70 \\ \hline 25 \end{array}$$

$$(1) - (2) \Rightarrow -x + y = -25 \rightarrow (4)$$

$$(3) + (4) \Rightarrow$$

$$x + y = 55$$

$$-x + y = -25$$

$$2y = 30$$

$$y = 15$$

Sub $y = 15$ in (3) we get

$$x + 15 = 55$$

$$x = 55 - 15$$

$$x = 40$$

$\therefore$ Age of Father = 40 years

Age of Son = 15 years

25. Let ~~the~~ $A(1, -7)$ and $B(6, 4)$ are divided by the point $(x, 0)$ lying on x-axis in ratio $m_1 : m_2$

$$(x, 0) = \left(\frac{m_1 x_2 + m_2 x_1}{m_1 + m_2}, \frac{m_1 y_2 + m_2 y_1}{m_1 + m_2} \right)$$

$$(6, 0) = \left(\frac{m_1 6 + m_2 1}{m_1 + m_2}, \frac{m_1 4 + m_2 (-7)}{m_1 + m_2} \right)$$

$$0 = \frac{4m_1 - 7m_2}{m_1 + m_2}$$

$$4m_1 - 7m_2 = 0$$

$$4m_1 = 7m_2$$

$$\boxed{\frac{m_1}{m_2} = \frac{7}{4} = 7:4}$$

$\therefore$ The ratio is 7:4

26.

$$\operatorname{cosec} \theta - \sin \theta = m$$

$$\frac{1}{\sin \theta} - \sin \theta = m$$

$$\frac{1 - \sin^2 \theta}{\sin \theta} = m$$

$$\frac{\cos^2 \theta}{\sin \theta} = m$$

$$\sec \theta - \cos \theta = n$$

$$\frac{1}{\cos \theta} - \cos \theta = n$$

$$\frac{1 - \cos^2 \theta}{\cos \theta} = n$$

$$\frac{\sin^2 \theta}{\cos \theta} = n$$

$$\text{LHS} := (m^2 n)^{2/3} + (mn^2)^{2/3}$$

$$= \left(\frac{\cos^2 \theta}{\sin \theta} \times \frac{\sin^2 \theta}{\cos \theta} \right)^{2/3} + \left(\frac{\cos^2 \theta}{\sin \theta} \times \frac{\sin^2 \theta}{\cos \theta} \right)^{2/3}$$

$$= \cos^{\frac{2}{3} \times 2/3} \theta + \sin^{\frac{2}{3} \times 2/3} \theta$$

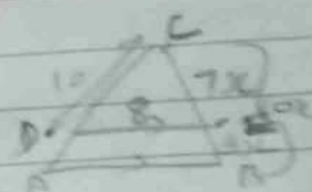
$$= \cos^2 \theta + \sin^2 \theta$$

$$= 1$$

$$\text{RHS} := 1$$

$$\text{LHS} = \text{RHS}$$

Hence, proved.



27.

(i)

$$\frac{\text{Side 1}}{\text{Side 2}} = \frac{\text{Perimeter 1}}{\text{Perimeter 2}}$$

$$\frac{10}{7} = \frac{10}{7}$$

(ii)

~~7 = 10~~ Let the distance between Hotel A and Hut 1 be x

$$\frac{7}{3} = \frac{10}{x} \quad (\text{By BPT})$$

$$7x = 30$$

$$x = \frac{30}{7} \text{ miles}$$

(iii) By AA similarity criteria,

$$\triangle ABC \sim \triangle DEC$$

$$\frac{DE}{AB} = \frac{CE}{EB}$$

$$\frac{8}{10} = \frac{7}{x}$$

$$7x = 80$$

$$x = \frac{80}{7} \text{ miles}$$

(iv)

$$AD = 5 + BC$$

$$10 = 5 + BC$$

$$BC = 5 \text{ miles}$$

$$\frac{EC}{BC} = \frac{7}{10} \Rightarrow \frac{EC}{5} = \frac{7}{10} \Rightarrow EC = \frac{7}{2} \text{ miles}$$

28.

$$\begin{aligned}
 (i) \quad Q &= \left(\frac{m_1 x_2 + m_2 x_1}{m_1 + m_2}, \frac{m_1 y_2 + m_2 y_1}{m_1 + m_2} \right) \\
 &= \left(\frac{1 \times 8 + 2 \times 2}{3}, \frac{1 \times 3 + 2 \times 5}{3} \right) \\
 &= \left(\frac{8 + 4}{3}, \frac{3 + 10}{3} \right) \\
 &= \left(\frac{12}{3}, \frac{13}{3} \right) \\
 Q &= \left(4, \frac{13}{3} \right)
 \end{aligned}$$

$$\begin{aligned}
 (ii) \quad PR &= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \\
 &= \sqrt{(8 - 2)^2 + (3 - 5)^2} \\
 &= \sqrt{(6)^2 + (-2)^2} \\
 &= \sqrt{36 + 4} \\
 &= \sqrt{40} \\
 &= 2\sqrt{10} \text{ units}
 \end{aligned}$$

$$\begin{array}{r}
 2 \overline{) 40} \\
 \underline{20} \\
 20 \\
 \underline{20} \\
 0
 \end{array}$$

(iii) Let the point be A

$$\begin{aligned}
 A &= \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right) = \left(\frac{2 + 4}{2}, \frac{\frac{13}{3} + 5}{2} \right) \\
 &= \left(\frac{6}{2}, \frac{\frac{14}{3} \times \frac{1}{2}}{\frac{1}{2}} \right) \\
 &= \left(3, \frac{14}{3} \right)
 \end{aligned}$$

$$\begin{array}{r}
 \frac{13+15}{3} \\
 = \frac{28}{3} \\
 = 9\frac{1}{3}
 \end{array}$$

(iv) Let the mid point of PR be B

$$B = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

$$= \left(\frac{8+2}{2}, \frac{5+3}{2} \right)$$

$$= \left(\frac{10}{2}, \frac{8}{2} \right)$$

$$= (5, 4)$$

29. distance = 100 km

Speed of car from A be x km/hr

Speed of car from B be y km/hr

$$100 = 50x - 5y$$

$$x - y = 20 \rightarrow (1)$$

$$100 = 2x + y \rightarrow (2)$$

$$(1) + (2) \Rightarrow$$

$$2x - y = 20$$

$$2x + y = 100$$

$$2x = 120$$

$$x = 60$$

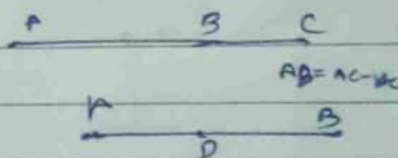
Sub $x = 60$ in (2) we get

$$60 + y = 100$$

$$y = 100 - 60$$

$$y = 40$$

$\therefore$ Speed of car from A = 60 km/hr
Speed of car from B = 40 km/hr



30.

Given :-

$$\angle ABD = \angle CAD = \angle PQR = 90^\circ$$

$$AB = x \text{ units}$$

$$CD = y \text{ units}$$

$$PQ = z \text{ units}$$

$$\text{To prove :- } \frac{1}{x} + \frac{1}{y} = \frac{1}{z}$$

Proof :-

$$\text{Let, } BQ = a \text{ units}$$

$$QD = b \text{ units}$$

In $\triangle BQP$ and $\triangle BDC$,

$$\angle B = \angle B \text{ (common)}$$

$$\angle Q = \angle D = 90^\circ$$

 $\therefore$ By AA similarity;

$$\triangle BQP \sim \triangle BDC$$

$$\frac{a}{a+b} = \frac{z}{y} \rightarrow (1)$$

Similarly,

$$\triangle PQR \sim \triangle DBA$$

$$\frac{b}{a+b} = \frac{z}{x} \rightarrow (2)$$

$$(1) + (2) \Rightarrow$$

$$\frac{a}{a+b} + \frac{b}{a+b} = \frac{z}{y} + \frac{z}{x}$$

$$\frac{a+b}{a+b} = 2 \left(\frac{1}{y} + \frac{1}{x} \right)$$

$$1 = 2 \left(\frac{1}{y} + \frac{1}{x} \right)$$

$$\frac{1}{2} = \frac{1}{y} + \frac{1}{x} \Rightarrow \frac{1}{x} + \frac{1}{y} = \frac{1}{z}$$

Hence, proved.

